

AI-Powered Analysis of Bodily Fluids for Early Disease Detection Using LLM Query Router

Introduction

Healthcare is witnessing transformation by AI with ever-evolving specifics technology. Traditional machine learning models tackle encoded medical information like blood work beautifully. On the other hand, large language models (LLMs) excel at general medicine operational queries. However, firstly combining these two approaches into one system is still an unresolved problem. This study analyses the hypothesis that a novel, deep-learning-routed AI classifier can boost diagnostic effectiveness by selecting the most appropriate query model. In other words, we are testing the viability of a multi-modal AI system that tackles both structured and unstructured medical data. An adaptive query classifier who’s going to determine which model to use is a core part of the system, but there are certain concerns how optimal model selection criteria accuracy, time, response, and computational cost are bound. Moreover, I explore the importance of self-improving AI systems with feedback loops that improve the decisions made for the aids of information self-sorting. To combat these challenges, we propose a system that incorporates three components, disease prediction model using static medical records, dynamic query classifier, and a simple general medical LLM that is widely used to be fed questions. By benchmarking this approach against existing query-handling methods, this research aims to advance AI-based clinical decision support systems to enable improved access and accuracy in health care diagnosis.

Background Research or Context

With the application of AI in the healthcare industry especially in disease diagnostics, there has been progress in building machine learning predictive models and large language models (LLMs) for retrieving medical knowledge. Both technologies work well on their own, however, the issue of merging them into a hybrid AI system with an intelligent dispatcher is still unresolved. This work focuses on the AI powered health care system gap that manages both structured and unstructured queries simultaneously within the system. The project builds on:
Medical AI Applications: Earlier works on using machine learning models for chronic kidney disease (CKD) and oral cancer detection.
Query Classification & Routing: Rule-based, ML based, and LLM based classifiers.
Adaptive Learning in AI Systems: Using feedback on the routing decisions over time.

Proposed Project

Objectives:

This research examines the potential of an AI-based query routing system to improve early disease diagnosis through the cross combination of structured medical data analysis and natural language processing. More specifically, the study looks at the effects, within query adaptive classification, on model selection, diagnostic accuracy, response time, and computational resources. The explorative work analyses the efficiency of medical query routing using different classification techniques, including rules, machine learning, and transformers. It also investigates the effectiveness of multi-modal AI systems on structured data (e.g. test results) and on unstructured medical queries as it relates to diagnosis accuracy. Furthermore, the research study also studies how feedback-driven query routing could be used to improve decision-making over time within AI-enabled healthcare through adaptive learning. Finally, the clinical effectiveness of the solution will be evaluated against other AI-based healthcare systems using real life medical data for benchmarking.

System Architecture:

Data model: It serves as the module for pre-processing the structured medical datasets to perform data cleaning, normalization, and feature extraction. It ensures compatibility between uploaded medical records and the disease prediction model.
Disease prediction model: A machine learning model trained on labeled medical datasets to assess the risk of disease based on structured inputs. The work will propose a suite of model architectures, for example, decision trees, logistic regression, neural networks, etc., to explore which works best for early disease detection.
Dynamic query classifier model (Router): The foci, this classifier determines what type of an incoming query should be routed through which channel. Structured data goes to the disease prediction model, whereas unstructured text-based queries are routed to a pre-trained LLM (i.e., ClinicalBERT, BioGPT). The research will explore different strategies, including rule-based methods and transformer-based classifiers, for optimum routing.
User Interaction Layer: The online interface for users to access and submit a query or medical test data. Accordingly, the system delivers AI-generated disease predictions or medically relevant answers. This system includes feedback on past queries and routing performance updates the classification model changes over time.

System Flow:

User Input: Users interact with the system via a web interface, entering queries or uploading datasets.
Query Classification: The dynamic query classifier determines whether the input is structured (e.g., test results) or unstructured (e.g., general medical inquiries).
Query Routing & Execution: Based on classification, structured data is processed by the disease prediction model, while unstructured queries are handled by a pre-trained LLM.
Feedback Mechanism: Performance metrics such as accuracy, response time, and computational cost are logged, enabling the system to refine its query-routing decisions over time.

Timeline

Milestone			Expected Completion
Project Definition	Document	(PDD)	28 February 2025
Disease Prediction Development	Model		31 March 2025
Query Classifier & Routing System			31 May 2025
Project Proposal submission			22 June 2025
Final Submission			22 August 2025

References

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